

1. Java Dequeue

In computer science, a double-ended queue (dequeue, often abbreviated to deque, pronounced deck) is an abstract data type that generalizes a queue, for which elements can be added to or removed from either the front (head) or back (tail).

Deque interfaces can be implemented using various types of collections such as `LinkedList` or `ArrayDeque` classes. For example, deque can be declared as:

```
Deque deque = new LinkedList<>();
```

or

```
Deque deque = new ArrayDeque<>();
```

You can find more details about Deque [here](#).

In this problem, you are given integers. You need to find the maximum number of unique integers among all the possible contiguous subarrays of size .

Note: Time limit is second for this problem.

Input Format

The first line of input contains two integers and : representing the total number of integers and the size of the subarray, respectively. The next line contains space separated integers.

Constraints

The numbers in the array will range between .

Output Format

Print the *maximum* number of unique integers among all possible contiguous subarrays of size .

Sample Input

```
6 3
```

5 3 5 2 3 2

Sample Output

3

Explanation

In the sample testcase, there are 4 subarrays of contiguous numbers.

- Has unique numbers.
- Has unique numbers.
- Has unique numbers.
- Has unique numbers.

In these subarrays, there are unique numbers, respectively. The maximum amount of unique numbers among all possible contiguous subarrays is .

PROGRAM

```
import java.util.*;
```

```
public class Solution {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int n = sc.nextInt();
```

```
        int m = sc.nextInt();
```

```
        Deque<Integer> deque = new ArrayDeque<>();
```

```
        Map<Integer, Integer> freq = new HashMap<>();
```

```
int maxUnique = 0;
```

```
for (int i = 0; i < n; i++) {
```

```
    int x = sc.nextInt();
```

```
    // Add element
```

```
    deque.addLast(x);
```

```
    if (freq.containsKey(x)) {
```

```
        freq.put(x, freq.get(x) + 1);
```

```
    } else {
```

```
        freq.put(x, 1);
```

```
    }
```

```
    // Remove oldest element
```

```
    if (deque.size() > m) {
```

```
        int removed = deque.removeFirst();
```

```
        freq.put(removed, freq.get(removed) - 1);
```

```
        if (freq.get(removed) == 0) {
```

```
            freq.remove(removed);
```

```
        }
```

```
    }
```

```
// Check unique elements
```

```
if (deque.size() == m) {
```

```
    maxUnique = Math.max(maxUnique, freq.size());
```

```
}
```

```
}
```

```
System.out.println(maxUnique);
```

```
sc.close();
```

```
}
```

```
}
```

OUTPUT

The screenshot shows a web browser window with the URL <https://www.hackerrank.com/challenges/java-dequeue/problem>. The page features a green banner with the text "Congratulations" and "You solved this challenge. Would you like to challenge your friends?" followed by social media icons and a "Next Challenge" button. Below the banner, a list of test cases is shown, all marked as successful. The first test case, "Test case 0", is expanded, displaying the following details:

- Compiler Message:** Success
- Input (stdin):**

```
1 6 3
2 5 3 5 2 3 2
```

A "Download" link is provided for the input.
- Expected Output:**

```
1 3
```

A "Download" link is provided for the expected output.

The browser's taskbar at the bottom shows the system date and time as 09:14 on 18-09-2026, along with various application icons and a search bar.